

Subject code: IIA4117	Subject name: Model Predictive Control	
Responsible subject teacher: Roshan Sharma	Campus: Porsgrunn	Faculty: TNM
Assignment given in WISEflow (date and time): 01.12.2020 Start time: 09:00		Submission time in WISEflow (date and time): 01.12.2020 End time: 12:30
No. of assignments: 3	No. of attachments: None	No. of pages incl. front page and attachments: 4

Aids and collaboration:

Permitted aids: None. Only Pen or pencil and blank paper sheet for writing answers.

	Yes	No
Is it an individual exam?	☒	☐
Is it allowed to collaborate with other persons?	☐	☒

Description of individual examination and illegal cooperation will be found at my.usn.no

Criteria for the answers:

Font type: Times New Roman Hand written solutions allowed	Font size: 12 Hand written solutions allowed	Line spacing: Single Hand written solutions allowed
No. of words (min/max): No limit	Maximum no. of pages excl. front page and attachments: No limit	

Other important information:

- Answer sheets written by hand are acceptable. The students should upload scanned copies of the answer sheets in pdf format, or take pictures of the answer sheet and upload them. All the uploaded documents **MUST** be clearly legible. It is the responsibility of the students to upload answer sheets that are clearly readable.
- It is strictly forbidden to communicate and collaborate, in any form, with other persons during the exam period.
- It is forbidden to use external documents and sources (like course lecture notes, course videos, books, articles, presentations, talks, other documents from the homepage of the course, etc.) during the exam period.
- This final written exam has a weight of 60% on the final grade. The remaining 40% weight on the final grade comes from the project work.
- The exam ending time also includes the time that the students have to use for scanning (or taking pictures) and uploading the answer sheets in Wiseflow. After kl. 12:30, it will not be possible to upload answer sheets in Wiseflow. Please allocate enough time for submission part of your answer sheets.

This exam counts 60% for the final grade

Problem 1

Gas lift is a widely used artificial lifting method for oil production from a reservoir. As an example, in Figure 1, three oil wells of a gas lifted oil field along with a gas compressor and a separator are shown.

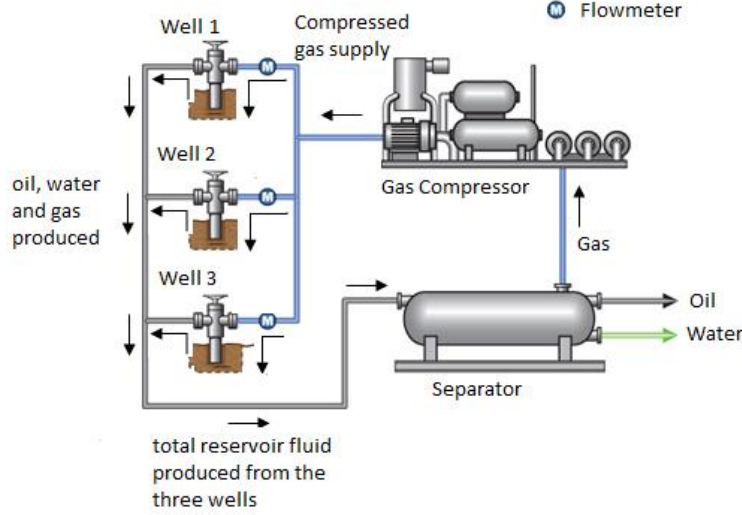


Figure 1: Gas lifted oil field with three oil wells.

In a gas lifted oil well, compressed gas is injected into the tubing/riser of the well. The compressed gas mixes with the reservoir fluid in the oil well and reduces its density. The less dense fluid in the well exerts lower hydrostatic pressure at the bottom and hence the bottom hole pressure of the well reduces. This causes more of the reservoir fluid (which is a mixture of oil, gas and water) to flow out of the well and hence enhanced oil recovery is achieved.

The gas to be injected to the wells is compressed by a compressor and then it is distributed to the three oil wells as shown in Figure 1. The reservoir fluid produced from each oil well is collected and transported to the separator where oil, water and gas are separated. The separated gas is recirculated to the compressor and then injected back into the wells.

The amount of oil recovered from each oil well ($w_{o,i}$) is a nonlinear function of the amount of gas injected into the well ($w_{g,i}$) and the well parameters θ_i . In discrete form, for any time k , it can be written as,

$$w_{o,i}^{k+1} = f(w_{o,i}^k, w_{g,i}^k, \theta_i). \quad (1)$$

Here, subscript i denotes the i^{th} oil well in the field with $i = 1, 2, 3$ and f is a nonlinear function. The gas lift performance curve shows the relationship between $w_{o,i}$ and $w_{g,i}$ as shown in Figure 2.

The well parameters θ_i are different for different oil wells. This means that the performance curve is different for each oil well. This further means that each of the oil wells produce different amount of oil for the same amount of gas injected into the well. Thus, the total amount of compressed gas should be distributed among the wells in such a way that the **total production** of oil from the field (sum of oil produced from each well) is maximized. At the same time, the cost of operating the compressor has to be minimized.

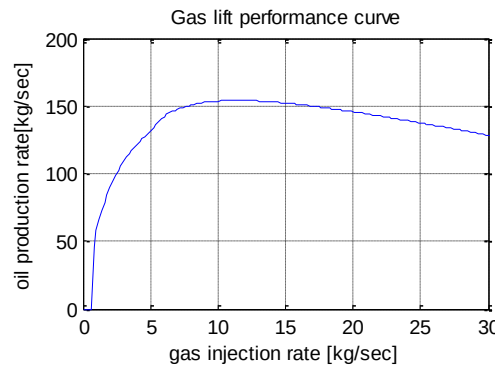


Figure 2: Gas lift performance curve for a typical oil well.

Consider C_{oil} to be the selling price of per unit oil produced from the field and C_{gas} to be the cost involved in compressing per unit gas. Assume $w_{o,i}$ and $w_{g,i}$ are expressed in $\frac{kg}{sec}$, and C_{oil} and C_{gas} are in $\frac{\$}{kg}$ respectively. As an expert in model based control, you would like to use a nonlinear MPC for optimal distributing of compressed gas to each oil well in the field. In addition, the following points should also be taken into consideration:

- i) The **total** amount of gas to be distributed in the three oil wells should not exceed the maximum capacity of the compressor ($w_{g,max}$).
- ii) The pressure in the separator for any time k (P_s^k) should not exceed $P_{s,max}$. Assume that separator pressure is a function of amount of gas produced from the wells which in turn is dependent on the amount of gas injected into the wells i.e.

$$P_s^k = g(w_{g,i}^k, \theta_i) \text{ for } i = 1, 2, 3. \quad (2)$$

Tasks:

The task is to distribute the compressed gas to the three oil wells of the field in such a way that the **total profit** obtained from the field (including all three oil wells) is maximized.

- a) (15%) Considering N as the prediction horizon length, write the formulation for a model predictive controller. Your formulation should include the objective function and the constraints. Also clearly state the variables that you are going to optimize.
- b) (10%) Assume that for a given value of $P_{s,max}$, the optimal control problem is infeasible. How will you handle it? What will be your assumption about the nature of the constraint associated with the variable P_s . Write the formulation for the optimal control problem with necessary adjustments in the objective function and the constraints.

Problem 2

Let us assume the nonlinear model of a process in continuous time is given as,

$$\frac{dx}{dt} = f(x, u, \theta) \quad x_0 \text{ is known,} \quad (3)$$

$$y = g(x, \theta). \quad (4)$$

Here, x, u, y and θ denotes the states, control inputs, outputs and parameters (assumed known) of the process. In addition, f and g represent some nonlinear functions for the state and measurement equations. Let us also assume that the process has n_x, n_y and n_u number of states, outputs and control inputs respectively. To control the outputs of the process, a linear MPC based on the following linearized model of the process is planned to be used.

$$\delta \dot{x} = A_c \delta x + B_c \delta u \quad \delta x_0 \text{ is known,} \quad (5)$$

$$\delta y = C_c \delta x. \quad (6)$$

Defining the operating point for states, control inputs and outputs to be x_{op}, u_{op} and y_{op} respectively, the deviation variables are, $\delta x = x - x_{op}$, $\delta u = u - u_{op}$ and $\delta y = y - y_{op}$. Let us assume that these operating points have non-zero values and are known. For a successful implementation of the linear MPC based on equations (5) and (6), the operating points should be properly handled, in particular, proper addition/subtraction of the operating points should be performed.

Now let us assume that you have been newly hired by the company as a control engineer, and that you have been assigned the task of designing a linear MPC for controlling the outputs of the process. Let us assume that during the development of the linear MPC based on equations (5) and (6), you often forget to add/subtract the operating points at proper places, and this has led to increasing frustrations, and your manager has started to

doubt your abilities. To come around this problem of yours, you decided to include the operating points in the formulation of the linear model itself, and thus you landed up with the following equivalent linear model of the process in discrete time as,

$$x_{k+1} = Ax_k + Bu_k + \Delta t \left(f(x_{op}, u_{op}, \theta) - (A_c x_{op} + B_c u_{op}) \right) \quad x_{k=0} \text{ is known,} \quad (7)$$

$$y_k = Cx_k + g(x_{op}, \theta) - C_c x_{op}. \quad (8)$$

Here, Δt is the sampling time.

Finally, the linear process model of equations (7) and (8) are to be used for designing an optimal controller for controlling the outputs to their desired setpoints. Let us consider a quadratic objective function J as,

$$J = \frac{1}{2} \sum_{k=1}^N e_k^T P_k e_k + u_{k-1}^T Q_{k-1} u_{k-1} + \Delta u_{k-1}^T R_{k-1} \Delta u_{k-1},$$

where,

$$e_k = r_k - y_k, \quad (9)$$

$$\Delta u_k = u_k - u_{k-1}.$$

Here, r_k is the reference for the output y_k and is considered to be known. Assume u_{-1} is known. $P_k > 0, Q_k \geq 0$ and $R_k \geq 0$ are the weighting matrices of appropriate sizes. We want to formulate an optimal control problem as a standard QP problem of the form,

$$\min_z \quad \frac{1}{2} z^T H z + c^T z,$$

s. t

$$A_\varepsilon z = b_\varepsilon, \quad (10)$$

$$A_i z \leq b_i,$$

$$z_L \leq z \leq z_H.$$

The vector of unknowns, z , is defined as,

$$z^T = [u_o^T, \dots, u_{N-1}^T, \Delta u_o^T, \dots, \Delta u_{N-1}^T, e_1^T, \dots, e_N^T, y_1^T, \dots, y_N^T, x_1^T, \dots, x_N^T]. \quad (11)$$

Tasks:

- (8%) Find the expression for matrix H and vector c .
- (15%) Find the expression for matrix A_ε and vector b_ε .

Problem 3

MPC problems can be computationally expensive. One simple way of handling computational time delay is by treating it as an output delay. Let us assume that the linear model of the plant being controlled using MPC is,

$$x_{k+1} = Ax_k + Bu_k \quad x_{k=0} \text{ is known,} \quad (12)$$

$$y_k = Cx_k \quad (13)$$

Let the sampling time for the model/process be 0.1 seconds. Let us now assume that the MPC algorithm unfortunately cannot be solved within the sampling time of 0.1 seconds at each time step. Let us assume that the maximum time required to obtain the solution of the linear MPC problem at any time step is 0.5 seconds.

Tasks:

- (7%) Write down the structure of the output delay model in a proper state space form. Clearly illustrate and write down the system matrices of the output delay model. You should show the contents of the system matrices of the output delay model.
- (5%) How would you use the output delay model together with the original state space model of the process in order to design a linear MPC?